Sakila Database Analysis

Uncovering critical insights from DVD rental data to optimize inventory, enhance customer loyalty, and drive revenue growth.

MINI PROJECT

GROUP 6

GROUP 6: MUHAMMAD HILMI BIN KAMIS | MATHIVANAN KALAIVANI | LIU XUQING | ROBIN CHIN

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Database Overview

1,000	599	16,044	2
Films	Customers	Rentals	Stores
Across 16 categories	Multiple cities	Transaction records	With staff & inventory

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INSIGHT 1

CRITICAL FINDING

Supply-Demand Mismatch

Significant misalignment between inventory supply and customer demand. High-demand categories like Action, Sci-Fi, and Drama are under-stocked, while Foreign films dominate inventory despite lower rental volumes.

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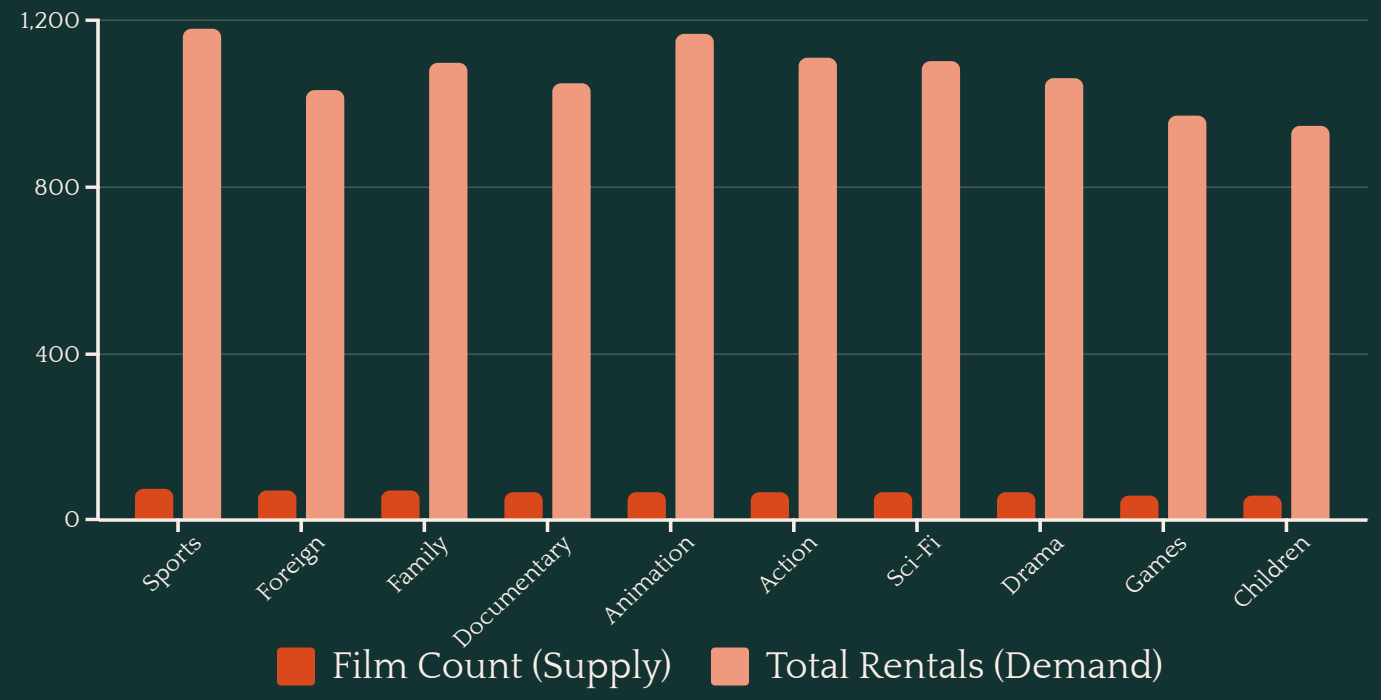
Inventory vs. Demand Analysis



Key Insights

Sports leads both supply (74 films) and demand (1,179 rentals), showing strong alignment. Animation generates high rental volume (1,166) with fewer films (66), indicating strong customer preference.

Foreign category has 73 films but ranks 8th in demand (1,033 rentals), representing tied-up capital in underperforming inventory.

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Revenue Opportunity

Action

1,112 rentals

Sci-Fi

1,101 rentals

Drama

1,060 rentals

Games

969 rentals

Children

945 rentals

Potential Impact

High-demand categories generated 5,000+ rentals. Proper stocking could increase total rental volume by **15-20%**.

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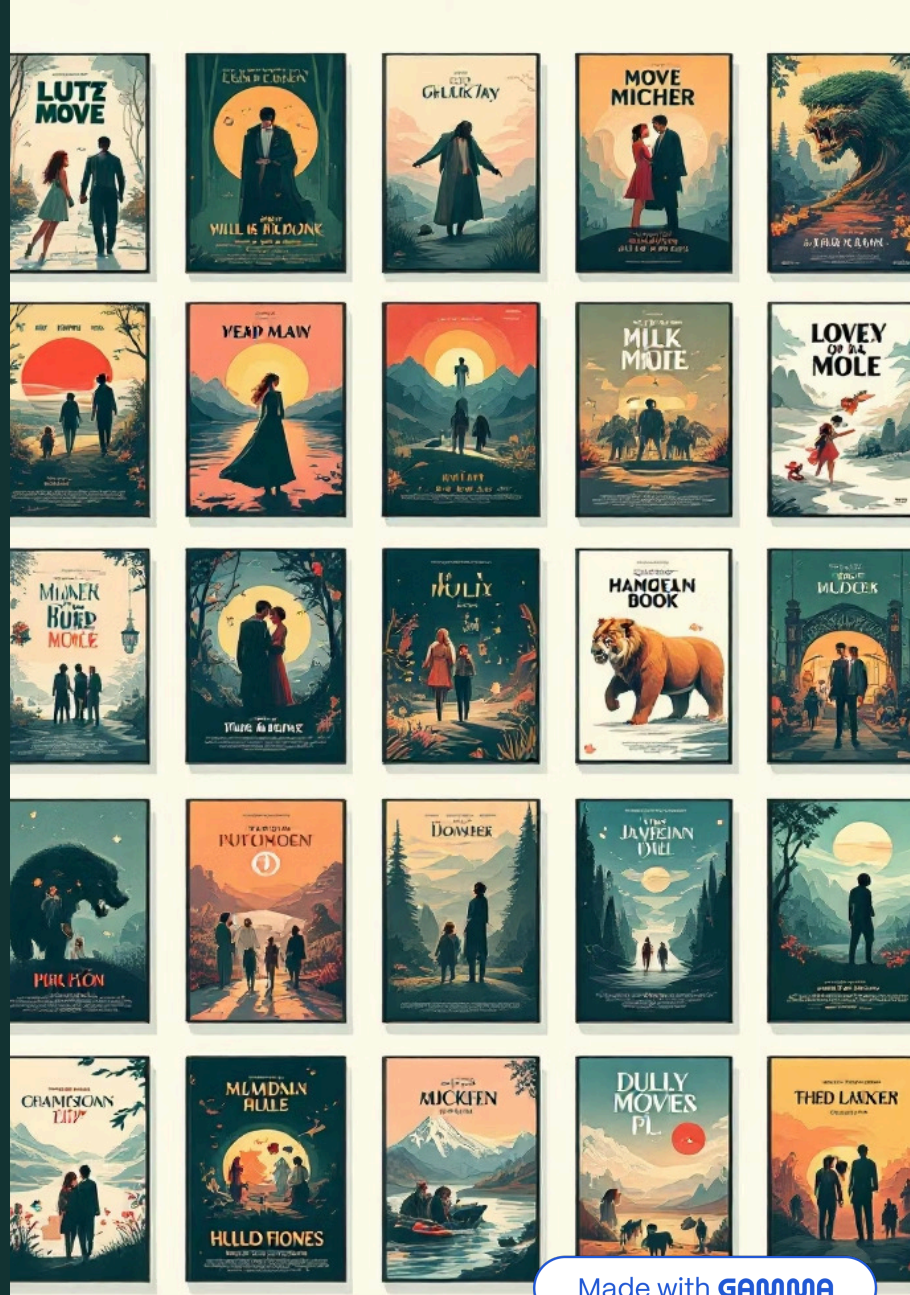
INSIGHT 2

Diverse Top Performers

Top 10 most rented films span 9 different categories, revealing diverse customer preferences and validating the need for broad catalog variety rather than genre concentration.



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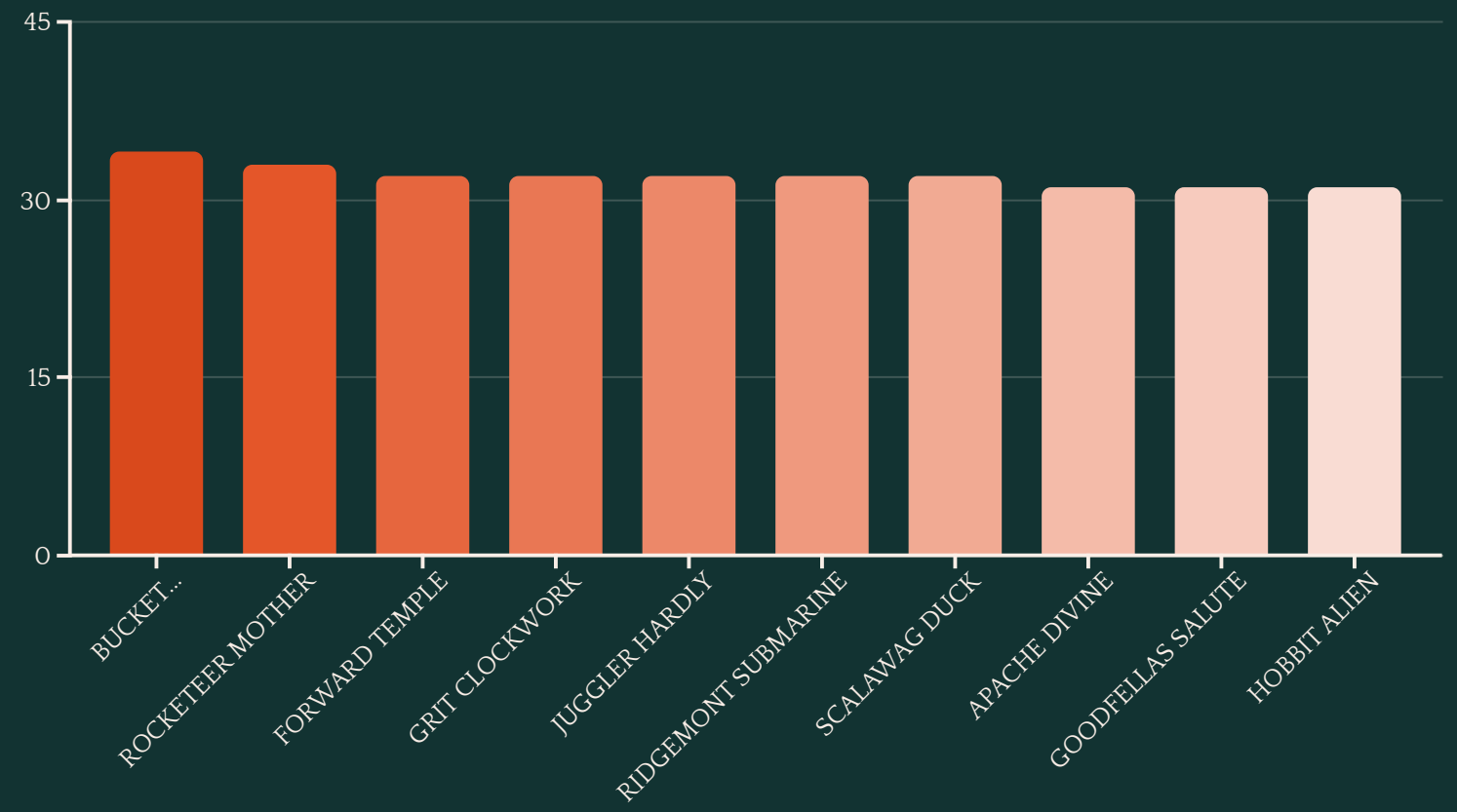
Most Frequently Rented Films



These top 10 most frequently rented films represent a diverse range of categories, highlighting varied customer tastes and significantly contributing to overall rental volume.

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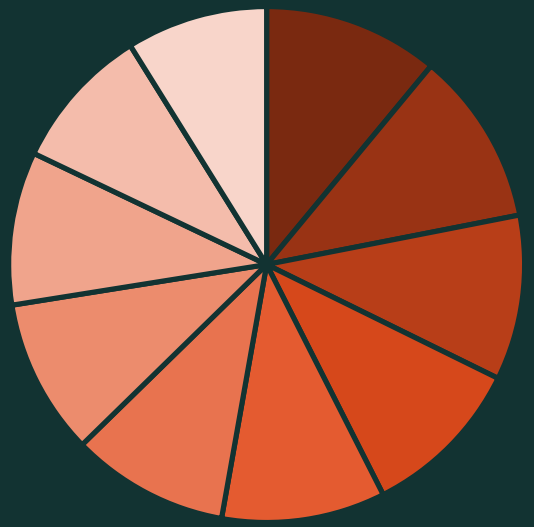
Category Market Share



Sports
Animation
Action
Sci-Fi
Family
Drama
Documentary
Foreign
Games
Children

Balanced Distribution

The pie chart illustrates a relatively balanced distribution of total rentals across all categories, indicating diverse customer preferences. **Sports** and **Animation** marginally lead the market share, followed closely by **Action** and **Sci-Fi**.

This balanced landscape suggests a broad appeal for various genres, rather than a heavy concentration in a few dominant categories. Understanding these proportions is crucial for strategic inventory planning and marketing efforts.

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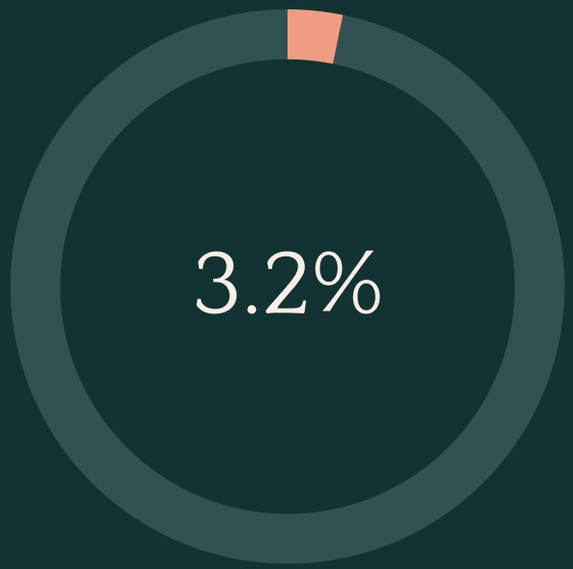
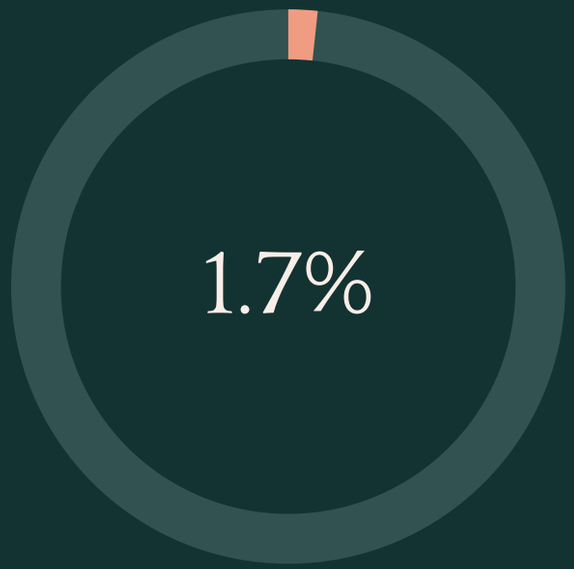
INSIGHT 3

VIP Customer Concentration



Revenue Share

From top 10 customers



Customer Base

Just 10 of 599 customers

High-Value Customers

Top 10 customers contributed **$1,917** in total revenue. Karl Seal leads at $221.55, with tight clustering among top performers ($175-$222 range).

Top customers represent **10-15x** the value of average customers, making retention critical.

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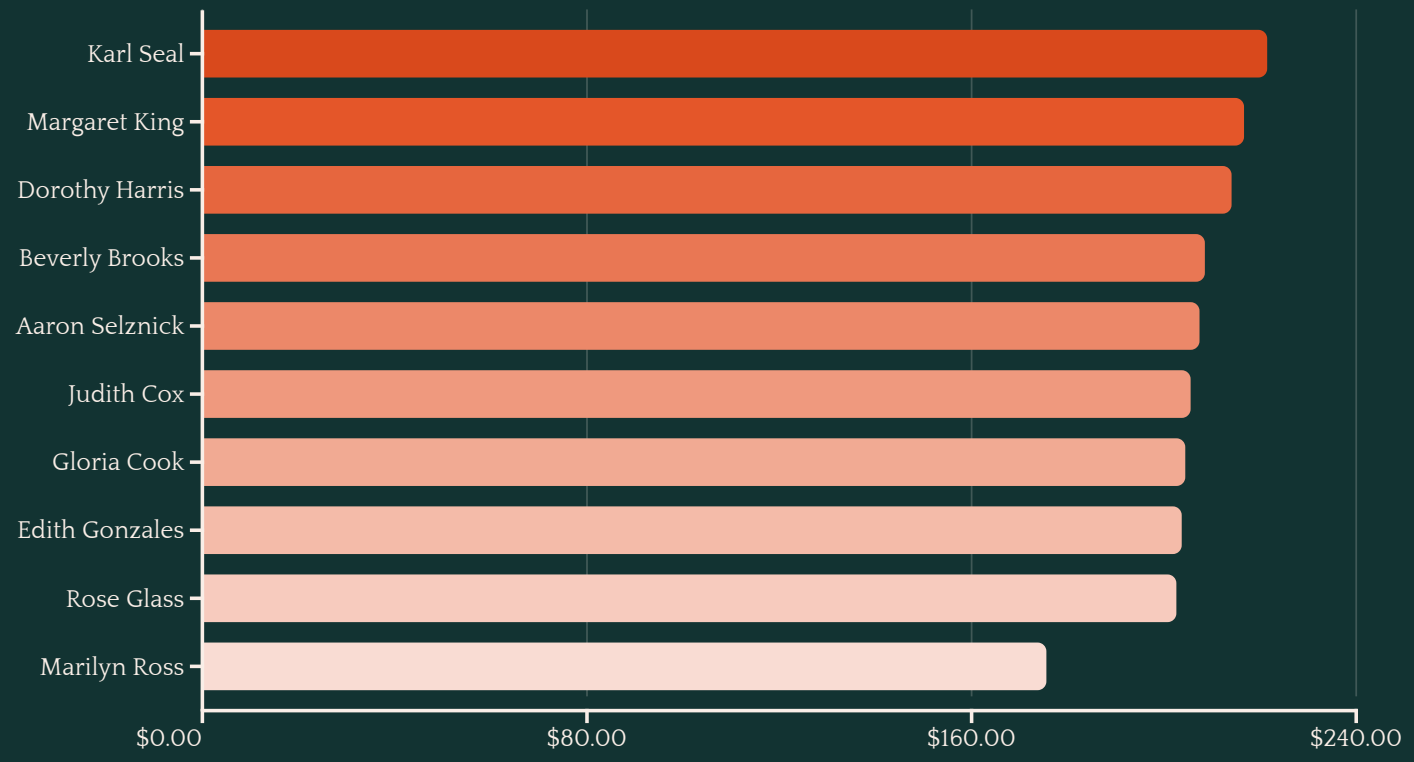
Top 10 Customers by Revenue



High-Value Customer Focus

This chart highlights the substantial revenue contributions from our top 10 customers, led by Karl Seal. These VIP individuals are critical to our business, demonstrating a clear concentration of revenue within a small segment of our customer base.

Targeted VIP retention programs and personalized engagement strategies are essential to nurture these relationships and maximize their long-term value.

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Strategic Recommendations

Optimize Inventory

Rebalance stock for high-demand categories (Action, Sci-Fi, Drama). Maintain 3-4 copies of top films per store.

VIP Loyalty Program

Create tiered membership with benefits: free late fees, priority holds, exclusive previews for top customers.

Monitor Performance

Establish regular reviews of category-level rental patterns and track VIP customer engagement.

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SQL Methodology

This section outlines the SQL queries employed for our analysis, categorizing them into three primary areas to uncover inventory insights, demand patterns, and customer value.

Top 5 Film Categories by Inventory Count

Objective: Identify which categories have the deepest catalog representation

Key steps: Join category → film_category → film tables, group by category, sort by film count

Output: Baseline supply metrics and over/under-represented categories

Top 10 Most Frequently Rented Films

Objective: Determine which specific films drive the most rental transactions

Key steps: Chain join film → inventory → rental, include category classification, group and sort by rental count

Output: Demand patterns and category diversity among top performers

Top 10 Customers by Total Spending

Objective: Identify highest-value customers based on cumulative payment amounts

Key steps: Join customer → payment, concatenate names, sum amounts, group and sort by spending

Output: Customer value hierarchy for targeted retention strategies

SQL Statements

Query 1: Top 5 Film Categories by Inventory Count

```
SELECT
c.name AS category_name,
COUNT(f.film_id) AS film_count
FROM category AS c
JOIN film_category AS fc
ON c.category_id = fc.category_id
JOIN film AS f
ON fc.film_id = f.film_id
GROUP BY c.category_id, c.name
ORDER BY film_count DESC
LIMIT 5;

Additional Analysis: Supply-Demand
To complete Insight 1, an additional query was executed to calculate rental volume by category:

-- Rental demand by category
SELECT
c.name AS category_name,
COUNT(r.rental_id) AS rental_count
FROM category AS c
JOIN film_category AS fc
ON c.category_id = fc.category_id
JOIN film AS f
ON fc.film_id = f.film_id
JOIN inventory AS i
ON f.film_id = i.film_id
JOIN rental AS r
ON i.inventory_id = r.inventory_id
GROUP BY c.category_id, c.name
ORDER BY rental_count DESC;
```

Query 2: Top 10 Most Frequently Rented Films

```
SELECT
f.title AS film_title,
c.name AS category_name,
COUNT(r.rental_id) AS rental_count
FROM film AS f
JOIN inventory AS i
ON f.film_id = i.film_id
JOIN rental AS r
ON i.inventory_id = r.inventory_id
JOIN film_category AS fc
ON f.film_id = fc.film_id
JOIN category AS c
ON fc.category_id = c.category_id
GROUP BY f.film_id, f.title, c.name
ORDER BY rental_count DESC
LIMIT 10;
```

Query 3: Top 10 Customers by Total Spending

```
SELECT
c.customer_id,
c.first_name || ' ' || c.last_name AS customer_name,
SUM(p.amount) AS total_spent
FROM customer AS c
JOIN payment AS p
ON c.customer_id = p.customer_id
GROUP BY c.customer_id, customer_name
ORDER BY total_spent DESC
LIMIT 10;
```

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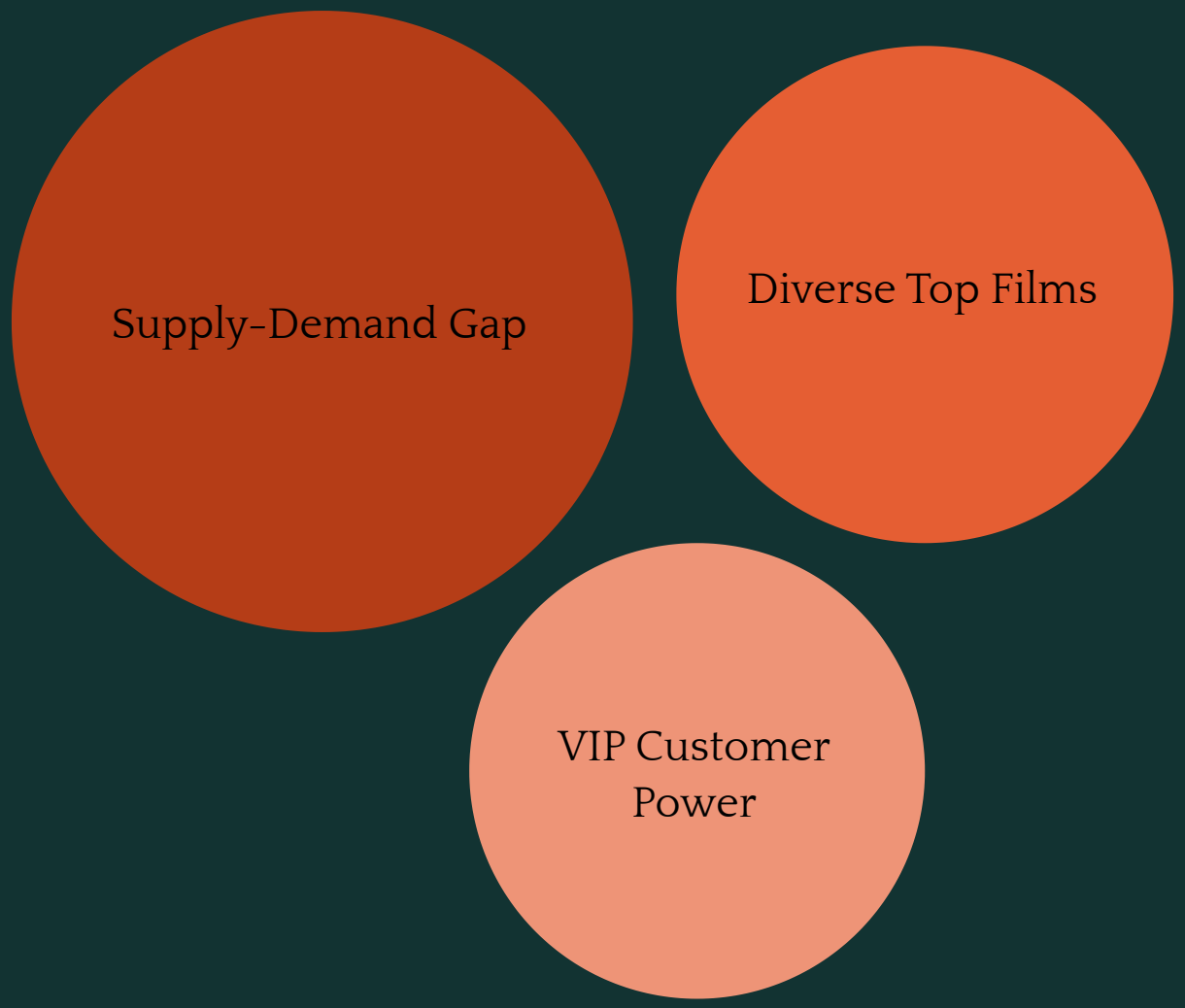
Key Takeaways



By addressing inventory inefficiencies, maintaining catalog diversity, and implementing targeted retention programs, Sakila can optimize operations and capture unrealized revenue potential.

Key Learning Points

- Data reveals hidden opportunities (15-20% revenue gap).
- Small changes, big impact (VIP programs and inventory rebalancing).
- Diversity is strength (9 categories in top 10 films).

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